



Next Generation of Policy Frameworks and Modeling

WEEK 2 • JUNE 15-19, 2026

Understanding Stock-Flow Dynamics and Steady States



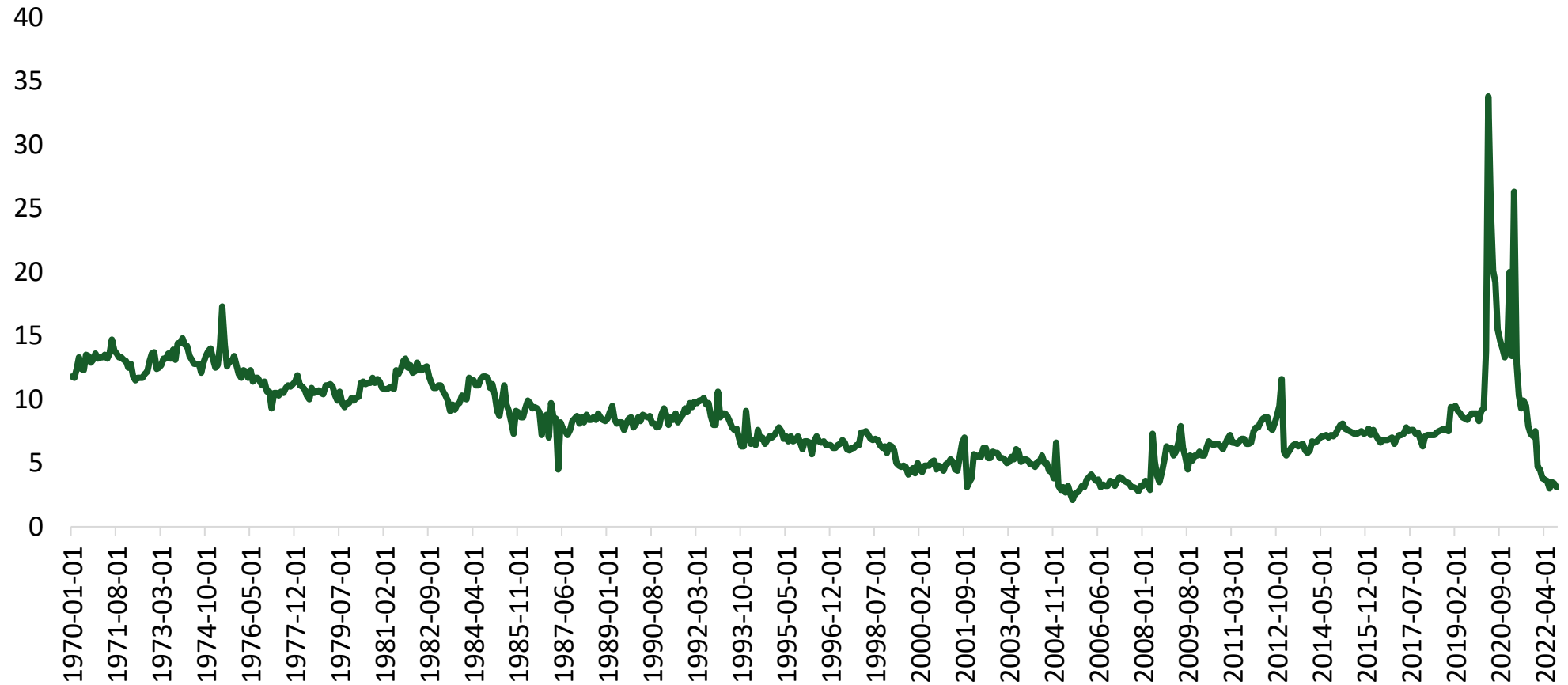
Outline

- Interpreting some trends
- Stock-Flow Dynamics in Steady-State Relationships
- Fiscal Policy, Government Debt and Deficits
- Capital Accumulation Processes (Private Sector Capital, Housing and Consumer Durables)



US Private Saving Rate

Personal Saving Rate

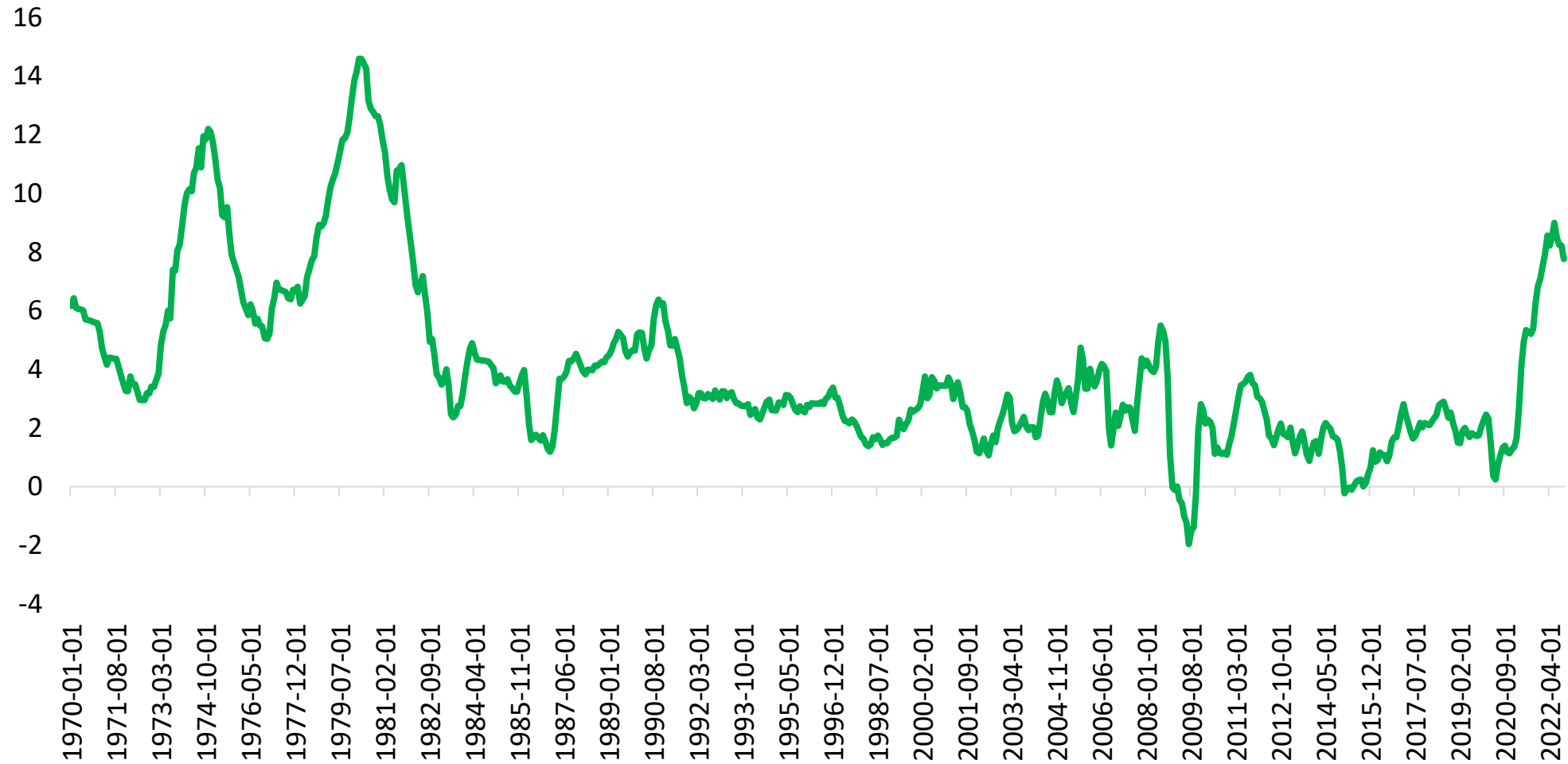


Source: FRED

<https://www.thebetterpolicyproject.org/>



US CPI Inflation

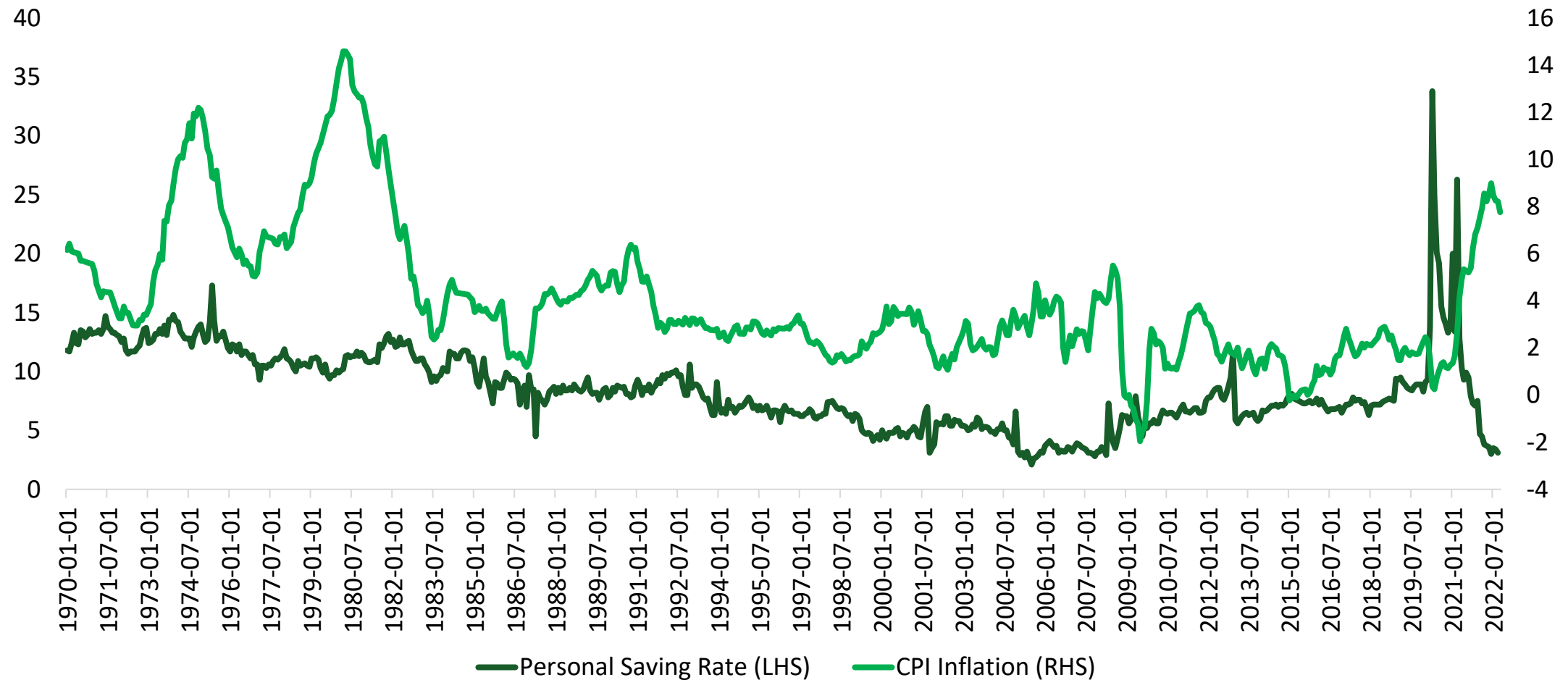


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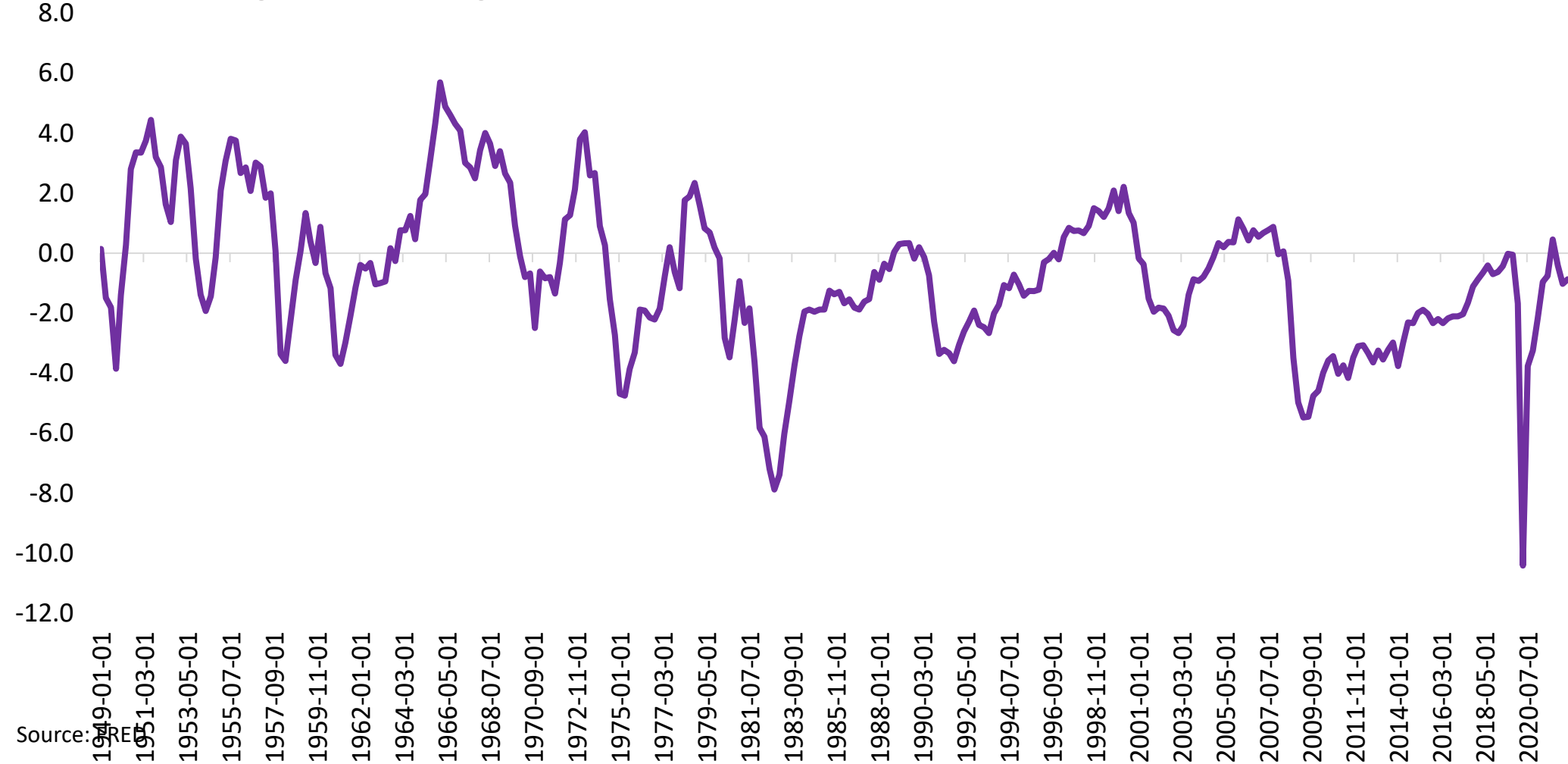
US Inflation and Private Saving Rate



Source: FRED



CBO Output Gap





Dynamic and Steady-State Models

DYNMOD - system of dynamic equations

$$\pi_t = 0.5\pi_{t+1} + 0.5\pi_{t-1} + 0.5Y_t + \varepsilon_{\pi t}$$

SSMOD - system of steady-state analogue equations

$$\pi_t = 0.5\pi_t + 0.5\pi_t + 0.5Y_T + \varepsilon_{\pi t}$$

~~$$\pi_t = 0.5\pi_t + 0.5\pi_t + 0.5Y_T + \varepsilon_{\pi t}$$~~

$$0.5Y_t = 0$$



$Y_t = 0$ in steady state

System of Steady-State Analogue Equations (SSMOD)...

...is maintained separately from the system of dynamic equations
(DYNMOD)

SSMOD can be used:

- As interpretive device for understanding long-run comparative statics
- For determining terminal conditions for simulations with the dynamic model



Examples of Stock-Flow Dynamics

Fiscal Deficit and Government Debt



Fiscal Deficit...

...is equal to the change in government debt:

$$DEF_t = D_t - D_{t-1}$$

where

DEF is fiscal deficit

D is government debt



Scaling by Nominal GDP

$$\frac{DEF_t}{YN_t} = \frac{D_t}{YN_t} - \frac{D_{t-1}}{YN_t}$$

where

YN is nominal GDP

$$G_t^{YN} = \frac{YN_t}{YN_{t-1}}, \quad g_t^{YN} = G_t^{YN} - 1$$

where

G_t^{YN} is growth rate of nominal GDP



Steady-State Growth Rate

$$G_{ss}^{YN} = \frac{YN_t}{YN_{t-1}} = \frac{DEF_t}{DEF_{t-1}} = \frac{D_t}{D_{t-1}}$$

where

G_{ss}^{YN} is the steady-state growth rate of all nominal GDP



Steady-State Growth Rate

- $\frac{D_t}{D_{t-1}} = G_{ss}^{YN} \Rightarrow D_{t-1} = \frac{D_t}{G_{ss}^{YN}}$

- $DEF_t = D_t - D_{t-1}$

$$DEF_t = D_t - \frac{D_t}{G_{ss}^{YN}}$$

$$DEF_t = \frac{D_t(G_{ss}^{YN} - 1)}{G_{ss}^{YN}} \Rightarrow \frac{DEF_t}{D_t} = \frac{(G_{ss}^{YN} - 1)}{G_{ss}^{YN}}$$



Nominal GDP Growth...

...is a combination of inflation and real growth

$$\Pi_t = \frac{P_t}{P_{t-1}} \quad \text{and} \quad G_t^Y = \frac{Y_t}{Y_{t-1}}$$

$$Y_N = P_t * Y_t$$

$$\pi_t = \Pi_t - 1$$

where

Π is inflation

P is price level

G is real growth rate



Nominal GDP Growth Rate

$$G_t^{YN} = \frac{Y_t * P_t}{Y_{t-1} * P_{t-1}} = G_t^Y * \Pi_t$$



Numerical Example

Assume

$$\Pi_t = 1.02 \quad \text{and} \quad G_t^Y = 1.02,$$

$$G_t^{YN} = \Pi_t * G_t^Y = 1.02 * 1.02 = 1.0404$$

$$\frac{DEF_t}{D_t} = \frac{(G_{ss}^{YN} - 1)}{G_{ss}^{YN}} \quad \Rightarrow \quad \frac{DEF_t}{YN_t} = \frac{(G_{ss}^{YN} - 1)}{G_{ss}^{YN}} * \frac{D_t}{YN_t}$$

$$\frac{DEF_t}{YN_t} = \frac{g_{ss}^{YN}}{(g_{ss}^{YN} + 1)} * \frac{D_t}{YN_t}$$



Numerical Example

$$\frac{DEF_t}{YN_t} = \frac{0.0404}{1.0404} * \frac{D_t}{YN_t} = 0.039 * \frac{D_t}{YN_t}$$

$\frac{D_t}{YN_t}$	$\frac{DEF_t}{YN_t}$ %
0	0
20	0.78
40	1.56
60	2.34
80	3.12
100	3.9

Fiscal Deficit in Terms of Primary Deficit and Interest Payments

$$DEF_t = DEF_t^P + INTPAY_t$$

where

DEF^P is primary budget deficit

$INTPAY$ are interest payments



Simple Model For Interest Payments

$$\text{INTPAY}_t = rn_t * D_{t-1}$$

$$rr_t = rn_t - \pi_{t+1}$$

where

rn is nominal interest rate

rr is real interest rate



Steady-State Growth Rate

$$G_{ss}^{YN} = \frac{YN_t}{YN_{t-1}} = \frac{DEF_t}{DEF_{t-1}} = \frac{D_t}{D_{t-1}}$$

where

G_{ss}^{YN} is the steady-state growth rate of all nominal GDP

Connection Between Primary Deficit and Government Debt

$$\text{DEF}_t = D_t - D_{t-1} \quad \text{and} \quad \text{DEF}_t = \text{DEF}_t^P + \text{INTPAY}_t$$



$$D_t - D_{t-1} = \text{DEF}_t^P + \text{INTPAY}_t$$

$$D_t - D_{t-1} = \text{DEF}_t^P + r_n * D_{t-1}$$

$$D_t = \text{DEF}_t^P + (r_n + 1) * D_{t-1}$$

Connection Between Primary Deficit and Government Debt

$$D_{t-1} = \frac{D_t}{G_{ss}^{YN}} \quad \Rightarrow \quad D_t = DEF_t^P + (rn_t + 1) * \frac{D_t}{G_{ss}^{YN}}$$

$$DEF_t^P = \frac{G_{ss}^{YN} * D_t - (rn_t + 1) * D_t}{G_{ss}^{YN}}$$

$$\frac{DEF_t^P}{YN_t} = \frac{(G_{ss}^{YN} - rn_t - 1)}{G_{ss}^{YN}} * \frac{D_t}{YN_t}$$

$$\frac{DEF_t^P}{YN_t} = - \frac{(rn_t - g_{ss}^{YN})}{(g_{ss}^{YN} + 1)} * \frac{D_t}{YN_t}$$



Numerical Example

Assume

$$rn_t=0.05 \Rightarrow \frac{DEF_t^P}{YN_t} = - \frac{(0.05-0.0404)}{1.0404} * \frac{D_t}{YN_t} = -0.0092 * \frac{D_t}{YN_t}$$

$\frac{D_t}{YN_t}$	$\frac{DEF_t^P}{YN_t} \%$
0	0
20	(-0.184)
40	(-0.368)
60	(-0.552)
80	(-0.736)
100	(-0.92)



Examples of Stock-Flow Dynamics

Current Account Balance and Net Foreign Assets



Current Account Balance...

...is equal to the change in net foreign assets:

$$CA_t = NFA_t - NFA_{t-1}$$

where

NFA is net foreign assets at the end of period



Current Account Identity (1)

$$\frac{CA_t}{YN_t} = \frac{NFA_t - NFA_{t-1}}{YN_t} \quad (1)$$

where

CA is current account balance

TB is trade balance

YN is the level of nominal gdp



Current Account Identity (2)

$$CA_t = TB_t + rn_t * NFA_{t-1}$$

where

rn is the net nominal interest rate



Scaling by Nominal GDP

This provides some illustrative calculations for the usual case in which nominal GDP is expanding over time

$$\frac{CA_t}{YN_t} = \frac{TB_t}{YN_t} + r n_t \frac{NFA_{t-1}}{YN_t} \quad (2)$$



Analogue Steady-State Equations

- Define g as steady-state net growth rate of all nominal variables, such that

$$NFA_t = NFA_{t-1} * (1 + g_t)$$

- Then for NFA_{t-1}

$$NFA_{t-1} = \frac{NFA_t}{(1 + g_t)}$$



Substituting in DYNMOD

- Equation (1)

$$\frac{CA_t}{YN_t} = \frac{NFA_t}{YN_t} * \frac{g_t}{(1 + g_t)} \quad (3)$$

- Equation (2)

$$\frac{CA_t}{YN_t} = \frac{TB_t}{YN_t} + \frac{NFA_t}{YN_t} * \frac{rn_t}{(1 + g_t)} \quad (4)$$



Steady-State Condition: No-Ponzi-Game

$$r > g$$

- If the real interest rates are less than the growth rate, governments could issue long-term debt with no apparent real costs.
- Governments reduce tax rates, issue debt instruments and allow growth in its tax base to eliminate the debt.



SSMOD as Interpretive Device

- Combining equations (3) and (4) we obtain an equation linking country's steady-state trade balance to its level of net foreign assets

$$\frac{TB_t}{YN_t} = - \frac{(rn_t - g_{ss})}{(1 + g_{ss})} * \frac{NFA_t}{YN_t} \quad (5)$$

- In long-run equilibrium, under the steady-state condition, when NFA/Y is stable, the country will receive (or pay) net income on its international asset (or liability) position that is a constant fraction of its GDP.



SSMOD as Interpretive Device

- Implications of equation (5) provide intuition for understanding the sensitivity of the long-run equilibrium level of the real exchange rate to shocks that affect the desired net foreign asset position of the economy.
- Increase in desired net foreign liabilities caused by an increase in government debt will require a larger trade surplus in order to finance the higher interest obligations. And this can be obtained only if the real exchange rate depreciates in the long run.



Numerical Examples

Under steady-state condition that $r > g$

%

r	g	NFA/Y	TB/Y
5	3	100	(-2)
5	3	(-100)	2



Government Debt, Net Foreign Liabilities and Real Exchange Rate

- Government debt in a particular country will influence real interest rates in the long run only if it affects global saving and investment and the level of the world real interest rate.
- The following 2 tables present estimates of the long-run effects of changes in the ratio of government debt to GDP for each of the major industrial countries

Increase of of Government Debt-to-GDP Ratio of 10 Percentage Points

	Net Foreign Liabilities ^{1,2}	Current Account Balance ^{1,3}	Trade Balance ^{1,3}	Real Competitiveness Index ²	Real Interest Rate ⁴
Full multicountry model results					
Canada	9.9	-0.5	0.2	-0.2	1
France	10.4	-0.5	0.3	-0.5	2
Germany	10.2	-0.5	0.2	-0.4	3
Italy	9.7	-0.5	0.2	-0.4	2
Japan	9.1	-0.4	0.2	-1.2	7
United Kingdom	9.3	-0.4	0.2	-0.3	2
United States	5.6	-0.3	0.2	-0.7	11

¹As a percent of nominal GDP.

²Percent deviation from baseline.

³Percentage point deviation from baseline.

⁴Basis point deviation from baseline.

Each row represents the simulation outcome for the increase of debt-to-GDP ratio of the country in the row.

Source: MULTIMOD Mark III : The Core Dynamic and Steady-State Models Douglas Laxton, Peter Isard, Hamid Faruquee, Eswar Prasad, and Bart Turtelboom



Interpreting the Results

- *In the first case*

Increase in debt leads to the almost same increase in net foreign liabilities. To service interest payments to foreigners trade balance surplus should be larger. This implies that the real exchange rate must depreciate in the long run.

- *In the second case*

Steady-state level of liabilities increases less. Shock has less effect on current account, real exchange rate. World real interest rate rises depending primarily on the size of the country in which the debt increase occurs.



Capital Accumulation Processes

Private Sector Capital, Housing, Consumer Durable Goods (Cars, Boats, Computers etc.)



Concept of the Balanced Growth Path

The capital stock and output should be growing at the same growth rate to have a balanced growth path. Let G^Y denote the gross growth rate in the steady state.

$$G^Y = \frac{Y_t}{Y_{t-1}} = 1 + g^Y$$

Most people usually think of growth rates as the percent change in the variable (g^Y), but it is much easier to do the math with gross growth rates. If GDP growth is at 2%, $G^Y = 1.02$ and $g^Y = 0.02$.



SS Stock-Flow Relationships (1)

Standard assumption is that the capital (K) depreciates at a constant rate δ .

$$K_t - K_{t-1} = I_t - \delta K_{t-1}$$

$K_t - K_{t-1}$ change in capital stock

I_t gross investment

δK_{t-1} depreciation



SS Stock-Flow Relationships (2)

We can also think of gross investment as a function of a sum of the change in capital stock and depreciation.

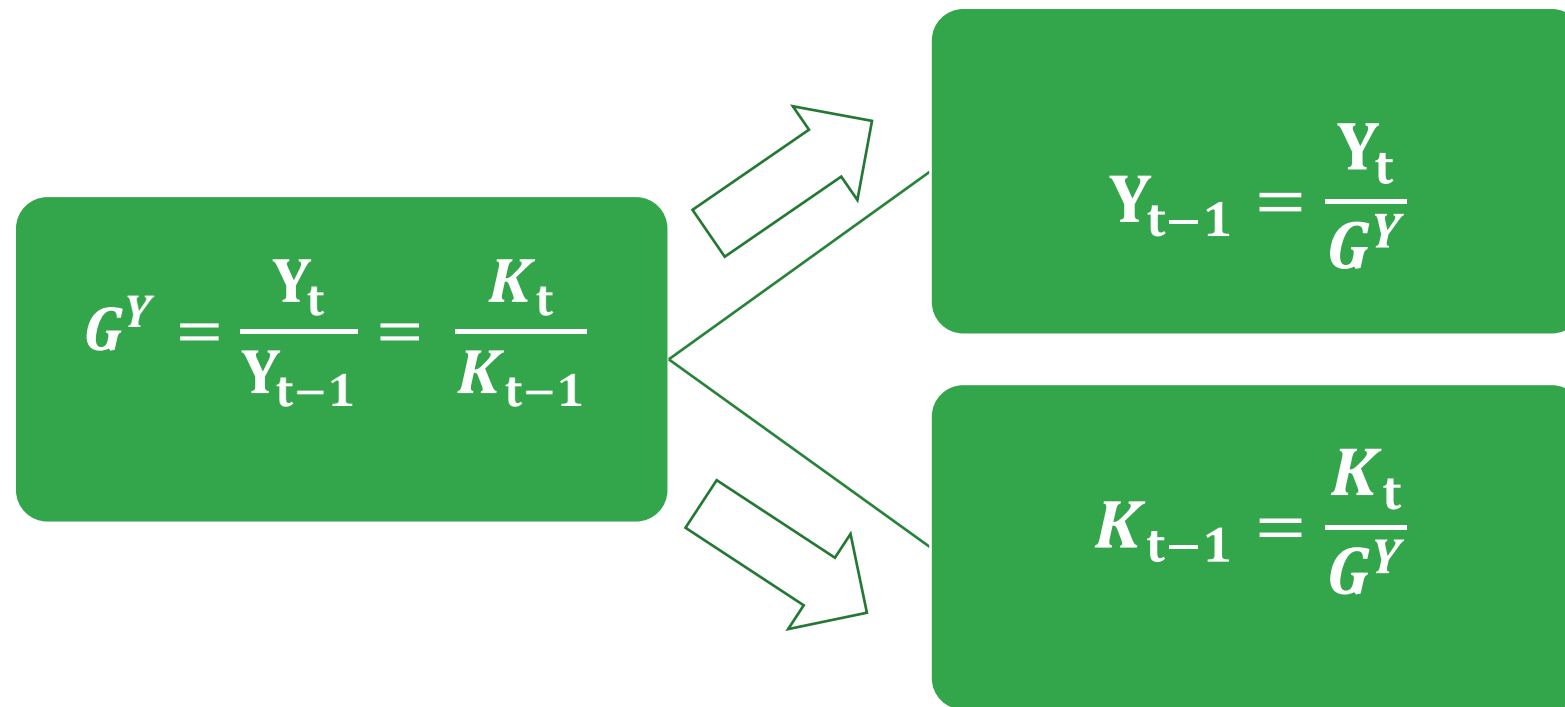
$$I_t = K_t - K_{t-1} + \delta K_{t-1}$$

I_t	gross investment
$K_t - K_{t-1}$	change in capital stock
δK_{t-1}	depreciation



SS Stock-Flow Relationships (3)

Levels of all real variables (Y_t , K_t etc.) must grow at the rate G_t^Y .





SS Stock-Flow Relationships (4)

$$I_t = K_t - K_{t-1} + \delta K_{t-1}$$

$$K_{t-1} = \frac{K_t}{G^Y} \Rightarrow I_t = K_t - \frac{K_t}{G^Y} + \delta \frac{K_t}{G^Y}$$

$$I_t = K_t \frac{(G^Y - (1 - \delta))}{G^Y}$$



SS Stock-Flow Relationships (5)

$$I_t = K_t \frac{G^Y - 1 + \delta}{G^Y}$$

$$g^Y = G^Y - 1 \Rightarrow I_t = \left(\frac{g^Y + \delta}{1 + g^Y} \right) K_t$$



Think about Housing (1)

Example: Depreciation rate $\delta = 0.02$ and $g^Y = 0.03$. Between 2002 and 2007 in the US average residential Investment to GDP ratio in real terms was 5.3%:

$$\frac{I_t}{Y_t} = \left(\frac{g^Y + \delta}{1 + g^Y} \right) \frac{K_t}{Y_t}$$
$$5.3 = \left(\frac{0.03 + 0.02}{1 + 0.03} \right) \frac{K_t}{Y_t}$$



Think about Housing (2)

This would imply that if the investment to GDP was fixed to 5.3% forever, than the capital-to-GDP ratio would converge to in the steady state:

$$\frac{K_t}{Y_t} = 5.3 \left(\frac{1+0.03}{0.03 + 0.02} \right) = 109.2\%$$

Now we have the relationship between the stocks and the flows that depend on the underlying growth rate and depreciation rate.

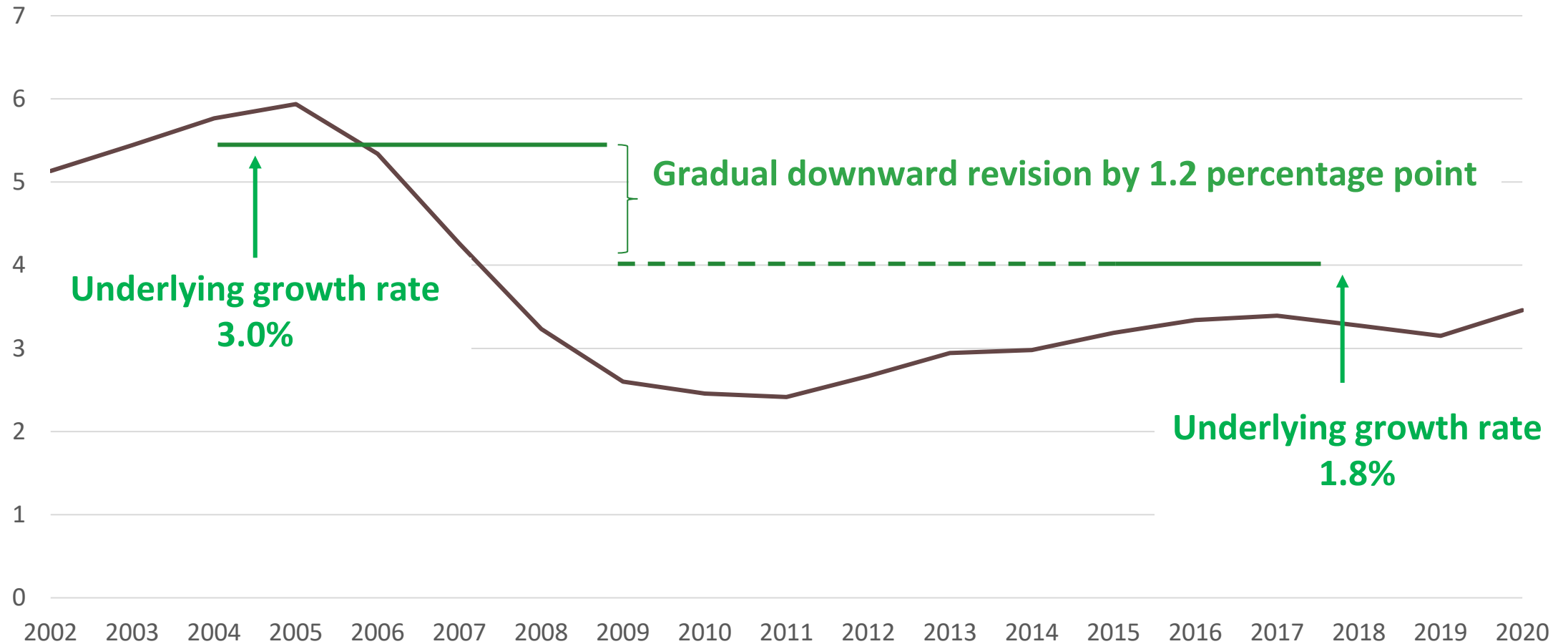


Think about Housing (3)

Before the global financial crisis people thought the sustainable rate of output growth was around 3.0% in the US. Today people think sustainable output growth is around 1.8%.

Sustainable output growth was revised down by 1.2 percentage points. We can account for a large chunk of the actual decline in the investment-to-GDP ratio with a decline in the steady-state investment-to-GDP ratio.

Private Fixed Investment, Residential, Share of GDP





Some Durables' Depreciation Rates

Type of asset	Rate of depreciation	Service life (years)
Durable goods owned by consumers		
Furniture, including mattresses and bedsprings	0.1179	14
Kitchen and other household appliances	0.15	11
China, glassware, tableware, and utensils	0.165	10
Other durable house furnishings	0.165	10
Video and audio products, computers and peripheral equipment, and musical instruments	0.1833	9
Jewelry and watches	0.15	11
Ophthalmic products and orthopedic appliances	0.275	6
Books and maps	0.165	10
Wheel goods, sports and photographic equipment, boats, and pleasure aircraft	0.165	10

Source: Bureau of Economic Analysis.

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